Mofe Omolade

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EDUCATION

University of British Columbia, *BASc in Computer Engineering* Expected 2029

Relevant Coursework: Circuit Analysis, Electricity & Magnetism, Linear Algebra, Embedded C, Engineering Design.

SKILLS

- Hardware & Protocols: PCBA Design, Power Electronics, CAN, SPI, I2C, UART, RS-485, STM32, ESP32, Arduino.
- Tools: Altium, KiCAD, LTspice, MATLAB, Git, VS Code, STM32 CubeIDE.
- Languages: C/C++, Python.
- Lab: Oscilloscopes, Logic Analyzers, DMM, Function Generators, Soldering, Circuit Debugging.

WORK EXPERIENCE

FulcrumAir, *Electronics Engineering Intern*, Calgary, AB May 2026 - Present

- Designed power distribution, communications breakout, and motor control PCBs for industrial robots in **Altium**.
- Designed an 8-layer motor control PCB in **Altium** around **Teensy 4.1**, IFX007T, MAX33040E, and LMR50410-Q1.
- Developed firmware for motor control, **CAN** communication, and linear actuator positioning feedback.
- Implemented a power hold-up system to trigger EEPROM data write within <5ms upon detecting voltage disconnect.

University of Calgary, *Electrical Engineering Research Assistant*, Calgary, AB July 2023 - August 2023

- Modelled and simulated Op-Amp oscillator circuits in **LTspice** to support research in amplifier-based oscillators.
- Contributed to academic projects across faculties and presented findings at symposiums and research boards.

DESIGN TEAM

UBC Rover, *Electrical Subteam*, Vancouver, BC September 2025 - Present

- Collaborated with chassis and software subteams to design a test rover for autonomous navigation training.
- Developed power architecture, integrated **UART** serial communication, and wrote low-level embedded firmware.
- Designed a powertrain prototype around the L298N full-bridge motor driver and **Nucleo F446RE** MCU.
- Implemented **C** firmware using **STM32** HAL for command parsing, motor feedback, and UART communication.

PROJECTS

Synchronous 24V-5V 2A Buck Converter

- Designed buck topology with dual N-channel MOSFETs, floating gate drives, and analog Type II loop compensation.
- Validated open-loop dynamics, small-signal Bode stability, and transient response in **LTspice** simulations.
- Performed 4-layer PCB layout in **Altium**, minimizing ground and switching loops to maintain signal integrity.

IMU Flight Attitude & Heading Reference System

- Developed **C** firmware for **ESP32** to parse high-rate IMU accelerometer and gyroscope streams via **I2C**.
- Reduced angular integration drift by applying a sensor fusion complementary filter and quaternion kinematics.
- Built a real-time aircraft Primary Flight Display **Python** GUI with pitch, roll, and yaw spatial telemetry.

Speedometer Heads-Up Display

- Designed a low-latency heads-up display using an Arduino Nano, GY-NEO6MV2 module, and SPI OLED display.
- Wrote **C++** firmware to parse **NMEA** data streams to compute velocity within 5% of the factory speedometer.
- Safely integrated with vehicle 14V input using a USB-C to DIN converter to supply the **Arduino** Vin.